BIM-Graphs: A Literature Review of Hybrid Building-Information Scene Graphs for Robot Navigation and Spatial Question Answering

Ashwin Nedungadi

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Abstract

3D scene graphs have become the dominant abstraction for hierarchical spatial reasoning in robotics, yet nearly all of them are built bottom-up from noisy onboard perception. Building Information Models (BIM), distributed as Industry Foundation Classes (IFC) files, already encode the metric, semantic and topological structure that such pipelines spend enormous effort estimating — and they do so before a robot has ever entered the building. This review surveys the two literatures that BIM-Graphs sits between: sensor-driven 3D scene graph construction, and BIM/IFC semantic modelling. We argue that the naive assumption underlying most BIM-to-graph work — that IFC's spatial decomposition tree is complete and trustworthy — is empirically false. An audit of 25 real building models (257 MB of IFC) finds zero IfcRelSpaceBoundary instances, only half of storeys containing any IfcSpace, and only 12% of elements assigned to a room. We conclude that the central research problem is not translation of IFC into a graph, but recovery of the topology that IFC omits. We then implement the recovery pipeline this implies and report what breaks: it produces 1,025 rooms (344 reconstructed) across the corpus and satisfies four necessary conditions universally, while exposing five pathologies of real IFC exports that the surveyed literature does not discuss. Because necessary conditions can falsify a graph but never confirm it, we identify ground-truth annotation as the binding constraint on further progress, and specify a correction-based protocol for obtaining it. An appendix reverse-engineers the design of the most relevant pipelines and identifies which of their components transfer to the BIM setting and under what conditions.

1 Introduction

A 3D scene graph represents an environment as a layered graph whose nodes are spatial concepts at increasing levels of abstraction — geometry, objects, places, rooms, buildings — and whose edges encode relations among them [1, 2]. The representation has proven unusually durable: it compresses dense geometry into something a planner can search [3], and something a language model can read without exhausting its context window [4].

The overwhelming majority of this work assumes the graph must be built by the robot. Kimera [2] and Hydra [5] derive rooms and places from an incrementally reconstructed mesh; S-Graphs [6, 7] co-optimises the graph with the robot's pose; ConceptGraphs [8] and HOV-SG [9] attach open-vocabulary features to segments recovered from RGB-D streams. This is the right design when no prior exists. For the built environment, however, a prior almost always does exist. Commercial buildings are designed in BIM authoring tools and delivered as IFC [10], a schema that already contains storeys, rooms, doors, walls, materials and exact world-frame geometry.

The apparent conclusion — that BIM makes scene graph construction trivial — is the misconception this review is written to dismantle. Exported IFC is a drawing-production artefact, not a topological one. The relationships a robot needs (which rooms are adjacent, which door connects which pair of rooms, which room contains which object) are precisely the relationships that authoring tools omit, because no drawing requires them. Section 6 quantifies this on real data.

Scope. We review four clusters: sensor-driven scene graphs (§2); open-vocabulary and language-grounded graphs (§3); BIM/IFC as a structured prior (§4); and space-boundary and room-recovery methods (§5). §6 synthesises these into the gap BIM-Graphs targets. Appendix A reverse-engineers the individual pipeline components.

2 Sensor-Driven 3D Scene Graphs

Armeni et al. [1] introduced the modern formulation: a four-layer graph over camera, object, room and building, constructed semi-automatically from panoramic scans of real buildings. The contribution was the structure, not the estimator, and that structure — objects nested in rooms nested in a building — has been inherited almost unchanged by everything since.

Kimera [2] made construction fully automatic from visual-inertial data, producing a five-layer Dynamic Scene Graph (mesh, objects and agents, places, rooms, building). Its decisive design choice was to derive the places layer from a Generalised Voronoi Diagram of free space, then segment rooms by detecting the narrow constrictions — doorways — that separate Voronoi regions. Hydra [5] made this real-time and, critically,

loop-closure aware: when the underlying pose graph is corrected, the entire hierarchy must deform consistently rather than be rebuilt. Mono-Hydra [11] showed the same architecture degrades gracefully to monocular input.

The S-Graphs line [6, 7, 12] takes a different position: rather than treating the hierarchy as a post-processing product of SLAM, it makes rooms and walls factors in the optimisation itself. A four-wall room imposes a geometric constraint that measurably improves pose estimates. S-Graphs 2.0 [12] exploits the hierarchy for computational gain, marginalising redundant keyframes within a room and reporting roughly an order-of-magnitude speed-up on multi-floor scenes.

Two works in this line are unusually close to our problem because they consume architectural priors directly. Shaheer et al. [13, 14] localise a robot by matching its situational graph against a graph extracted from architectural plans, and Ndulue et al. [15] replace the combinatorial matcher with a learned, end-to-end differentiable one, reporting orders-of-magnitude speed-ups at competitive matching quality. These establish BIM-derived graphs as a registration target; they do not address how to build a reliable graph when the BIM's own topology is missing.

3 Open-Vocabulary and Language-Grounded Graphs

Closed label sets are the binding constraint on the systems above. 3DSSG [16] and SceneGraphFusion [17] established learned semantic scene graph prediction over fixed vocabularies; Open3DSG [18] removed the fixed relation set by distilling 2D vision-language features into a 3D graph encoder and querying a grounded LLM for open-set predicates. ConceptGraphs [8] takes the pragmatic route — class-agnostic segmentation, CLIP embeddings per segment, LLM-inferred inter-object relations — and HOV-SG [9] adds the hierarchy back, building floor/room/object levels with open-vocabulary features at each and a cross-floor Voronoi graph for traversal, reporting a 75% memory reduction against dense feature maps. Clio [19] argues the correct granularity is task-dependent and prunes the graph against a task list using an information-bottleneck criterion — directly relevant to us, since a BIM contains vastly more detail than any single robot task needs.

Recent work pushes toward function rather than appearance: functional scene graphs [20] and TB-HSU [21] model affordances and interactions, which is what the BIM-Graphs design specification calls "semantic injection." KeySG [22] avoids explicit pairwise edges entirely, storing keyframe-grounded multi-level features and retrieving through a RAG interface — evidence that the edge set need not be materialised if retrieval is good enough.

On the reasoning side, SayPlan [4] demonstrated the property that motivates our downstream task: an LLM can plan over a building-scale graph if permitted to collapse and expand it, exploring a small subgraph at a time instead of ingesting the whole scene. 3DGraphLLM [23], SG-Nav [24] and MSGNav [25] extend graph-conditioned reasoning to grounding and zero-shot navigation, and multi-robot hierarchical planning [26] scales it across agents.

4 BIM and IFC as a Structured Prior

The AEC community arrived at graph representations independently and earlier. ifcOWL [27] mechanically maps the IFC EXPRESS schema to OWL, preserving everything at the cost of a verbose, deeply reified graph that is awkward to query. The Building Topology Ontology (BOT) [28] responds with a deliberately minimal core — zones, storeys, spaces, elements and adjacency — and is far closer to what a robot actually needs. The lesson transfers directly: completeness and queryability are in tension, and robotics wants the queryable end.

Recent work couples these graphs to language models. Iranmanesh et al. [29] convert IFC to a knowledge graph and use an LLM to emit Cypher, retrieving grounded facts rather than trusting parametric memory — the Graph-RAG pattern our specification adopts. Text2BIM [30] inverts the direction, generating models from text via a multi-agent framework.

Most directly, BIM-Edit [31] benchmarks LLMs on natural-language editing of IFC across 324 tasks, scoring geometric accuracy, semantic validity and topological consistency. Its headline result is sobering and strongly motivates the present work: the best model reaches only 49.5% average score, and none fully solves more than 3.4% of tasks, with topological instructions the weakest category. Raw IFC is evidently a poor interface for language models; an explicit, verified topological graph is the natural remedy.

For navigation specifically, Zhu et al. [32] build a semantics-based connectivity graph via IFC-Graph, connecting neighbouring spaces through doors for indoor pathfinding. Gopee et al. [33] export navigation maps, simulation environments and waypoints from IFC to remove the manual pre-mapping step, and Karimi et al. [34] formalise the exchange ontologically. Vega Torres et al. [35] close the loop by localising a 2D LiDAR against a BIM-derived map for lifelong navigation. Pix2G [36] runs the opposite direction in real time on a NASA JPL NeBula-Spot, producing BIM-like top-down maps and a multi-layer graph on CPU alone.

Crucially, every one of these BIM-to-graph works presumes a well-formed spatial hierarchy, typically that IfcSpace entities exist, are correctly nested under storeys, and are related to bounding elements. Our audit shows this presumption fails on realistic exports.

Property of the 25-model corpus	Value
Total IFC volume	257 MB
Storeys / rooms / doors	146 / 681 / 951
IfcRelSpaceBoundary (any level)	0
Storeys containing ≥ 1 IfcSpace	74 / 146 (51%)
Models with <50% storey coverage	10 / 25
Models with no IfcSpace at all	2 / 25
Models with no storey containment at all	1 / 25
Elements assigned to a room	2,001
Elements assigned only to a storey	15,349 (88%)
Rooms carrying a functional label	671 / 681 (98.5%)

Table 1: Schema-level audit of the 25-model test corpus. Not one file in 257 MB of IFC provides a single space boundary, and half of all storeys contain no room whatsoever — yet almost every room that is present is semantically labelled. IFC is a strong source of semantics and a weak source of topology.

5 Recovering Spaces and Boundaries

The AEC literature has, in fact, long known that IFC space data is unreliable — it surfaced first in building-energy simulation, which needs second-level space boundaries for thermal analysis. Lilis et al. [37] generate that topology from raw IFC geometry, explicitly handling clash errors, malformed space volumes and surface-orientation errors. Fichter et al. [38] note plainly that second-level boundaries in IFC are "often missing or incorrect" and that many delivered files lack IfcSpace entirely, generating the geometry themselves. This literature is largely unknown in robotics, yet it solves precisely the problem that blocks BIM-driven scene graphs.

Where spaces are absent altogether, room recovery becomes a segmentation problem. Bormann et al. [39] survey and benchmark the four families — morphological, distance-transform, Voronoi and feature-based — on occupancy grids, and Voronoi methods remain the default (and are what Kimera and Hydra adopt internally). Ochmann et al. [40] reconstruct volumetric, parametric building models from point clouds, casting room labelling as graph-cut energy minimisation over a line arrangement of wall centre-lines. Their motivation is the same "double-sided wall" pathology named in our design specification: a wall is a solid with two faces, and treating it as a zero-thickness surface corrupts adjacency. Kim and Min [41] add an LLM layer on top of geometric segmentation to assign functional room labels.

6 Synthesis and Positioning

Table 1 reports our audit. Three findings organise the project.

(1) Space boundaries do not exist. Across 257 MB of IFC there are zero IfcRelSpaceBoundary instances of any level. The schema-sanctioned route from a room to its bounding walls, and from a door to the pair of rooms it joins, is simply unavailable. Every wall–room and door–room edge must be recovered geometrically. This validates transplanting the energy-minimisation and boundary-generation machinery of [37, 40] into a robotics pipeline.

(2) Room coverage is partial and unreliable. Only 74 of 146 storeys (51%) contain any IfcSpace; upper storeys and service levels are routinely unmodelled, 10 of 25 models fall below half coverage, and two contain no room at all. A pipeline that traverses IfcBuildingStorey → IfcSpace — as our own design specification currently proposes — silently produces an empty graph for the majority of the building. Geometric room inference is therefore not an optional fallback but a required stage. Encouragingly, where spaces are present they are well labelled (671/681 carry a functional LongName such as kitchen or staircase), so the schema is a strong source of semantics even when it is a weak source of topology — a division of labour that shapes our architecture.

(3) Containment is nearly absent. Only 2,001 elements are assigned to a room against 15,349 to a storey alone, and one model states neither. Object-in-room edges — the backbone of both navigation and spatial QA — must be computed, not read. A further hazard compounds this: units are not reliable either. One model declares no IfcUnitAssignment whatsoever while its coordinates are in centimetres, so a parser that trusts the declared unit silently produces rooms 10^4 times too large.

Positioning. BIM-Graphs therefore differs from sensor-driven pipelines [5, 9] in that it never estimates geometry, and from existing BIM-to-graph work [32, 33] in that it does not assume the topology it needs is present. It is best understood as a topology-recovery system: it inherits room segmentation from the robotics literature [39], boundary generation from building simulation [37], hierarchy from Kimera and Hydra [2, 5], the collapse/expand query interface from SayPlan [4], and the Graph-RAG retrieval pattern from [29]. The value proposition — a graph that is metrically exact by construction, with no SLAM drift and no hallucinated objects, yet with topology guaranteed complete rather than inherited — is, to our knowledge, unoccupied. Its natural evaluation targets are robot navigation and the spatial reasoning tasks on which raw-IFC LLMs demonstrably fail [31].

7 Empirical Study: Building the Recovery Pipeline

To test whether the gap identified above is real and tractable, we implemented the recovery pipeline of Figure 2 and ran it over the full 25-model corpus. This

	Necessary condition	Pass
R2	every door joins exactly two rooms	25/25
R3	every room has one storey parent	25/25
R4	areas and storey heights are physical	25/25
R6	elements lie inside their assigned room	25/25
R1a	intra-storey connectivity ≥ 0.90	10/25
R1b	storeys mutually reachable	10/24
R5	element–room coverage ≥ 0.50	18/25

Table 2: Validation over the 25-model corpus; mean intra-storey connectivity is 0.789. R1b is not applicable to one model, which contains no vertical circulation of any kind. These are necessary conditions, not accuracy: they can falsify a graph but cannot confirm that a recovered corridor corresponds to a real one.

section reports what actually broke. We include it because the failure modes are not incidental engineering detail: several are properties of real IFC exports that any future BIM-to-graph system will meet, and none are discussed in the literature surveyed above.

7.1 Results

All 25 models process end-to-end (median ∼50 s per model, CPU only), yielding 1,025 rooms — 681 read from IFC and 344 reconstructed geometrically — with 678 resolved door connections and 10,796 catalogued elements. Because no ground-truth scene graph exists for these buildings, which is the premise of this work, we validate against necessary conditions that any correct building graph must satisfy rather than against a reference. Four hold universally; connectivity does not (Table 2).

7.2 Pathologies of real IFC exports

Five defects had to be solved before any topology could be recovered. Each is invisible in the schema and fatal if unhandled.

(P1) Units are not trustworthy. One model declares no IfcUnitAssignment on its IfcProject at all while its coordinates are in centimetres; geometry kernels then default to metres and every room emerges 10^4 times too large. Because a declared unit may be absent or wrong, we recover it empirically — floor-to-floor height is physically bounded to roughly 2–6 m, which identifies the true unit to within a decimal power. A related trap: IfcOpenShell already converts geometry to metres, so applying the file's unit scale a second time silently shrinks millimetre models by 10^3.

(P2) Sibling storeys share GUID prefixes. Revit assigns adjacent storeys near-identical GlobalIds (e.g. 1qw8GM0X…es, …ez, …ec), so keying derived nodes on a truncated GUID silently merges rooms across floors. This failure is especially treacherous because it raises apparent connectivity while destroying the hierarchy — a metric improving for a bad reason.

(P3) Doors are barriers first, connections second. IFC models doors as solids filling their openings. Rasterising them closed is what makes the connected components of free space separate cleanly into rooms; door connectivity is restored afterwards by probing along each door's placement normal. Under world-coordinate tessellation the shape transform is reported as identity, so that normal must be read from the placement, not the shape.

(P4) Not every transition is a door. Open stairs, archways and open-plan thresholds carry no IfcDoor. Relying on doors alone left the entire ground floor of one model unreachable from the storeys above. We instead test whether the strip of space between two adjacent rooms is actually filled by wall solids; a mostly empty gap is a passage. This single addition moved that model from 0.52 to 1.00 reachability.

(P5) Rooms are drawn for lettable area, not circulation. Where a doorway is open at slice height, the free-space component spans the corridor and the rooms off it. Discarding such a component because it overlaps stated rooms deletes precisely the corridor that makes those rooms mutually reachable; subtracting the stated rooms instead leaves the corridor behind. On one model this raised resolved doors from 14 to 32 and reachability from 0.92 to 1.00.

7.3 What the residual failures indicate

The remaining failures are concentrated and informative. Intra-storey connectivity (R1a) fails predominantly on open-plan, glazed buildings whose enclosure is curtain wall rather than IfcWall — one model holds 2,098 IfcPlate against 51 IfcWall — where there are few partitions to recover at all. This is a genuine limitation of occupancy-slice segmentation inherited from [39], not a tuning artefact.

Several R1b failures are not attributable to the method: one model contains zero IfcStair, IfcRamp and IfcTransportElement across seven storeys, each of which is perfectly connected internally but cannot be joined by any means. We record such cases as not-applicable rather than failed, since scoring a recovery method on topology the source never contained measures the wrong thing.

Diagnosis depended on inspecting what the isolated rooms were, not the aggregate. They proved to be bathrooms, hallways, offices and kitchens of median area 13 m^2; isolated hallways are self-evidently wrong, and that is what pointed to P5. The aggregate alone would merely have suggested a threshold to tune.

7.4 Consequence: ground truth is the bottleneck

Necessary conditions can falsify a graph but never confirm one. Nothing in Table 2 establishes that a recovered corridor corresponds to a real one, and precision/recall over rooms and connectivity edges cannot be computed without annotation. A hand-annotated subset is therefore the critical next step and the principal obstacle to any quantitative claim; we are building a correction-based annotation tool (Appendix B) for that purpose.

References

[1] I. Armeni, Z.-Y. He, J. Gwak, A. R. Zamir, M. Fischer, J. Malik, and S. Savarese, "3d scene graph: A structure for unified semantics, 3d space, and camera," in Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 5664–5673, 2019.

[2] A. Rosinol, A. Violette, M. Abate, N. Hughes, Y. Chang, J. Shi, A. Gupta, and L. Carlone, "Kimera: From SLAM to spatial perception with 3D dynamic scene graphs," The International Journal of Robotics Research, vol. 40, no. 12-14, pp. 1510–1546, 2021.

[3] A. Ray, C. Bradley, L. Carlone, and N. Roy, "Task and motion planning in hierarchical 3D scene graphs," 2024.

[4] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suenderhauf, "SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning," 2023. CoRL 2023.

[5] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," 2022. Robotics: Science and Systems (RSS).

[6] H. Bavle, J. L. Sanchez-Lopez, M. Shaheer, J. Civera, and H. Voos, "Situational graphs for robot navigation in structured indoor environments," 2022. IEEE RA-L.

[7] H. Bavle, J. L. Sanchez-Lopez, M. Shaheer, J. Civera, and H. Voos, "S-Graphs+: Real-time localization and mapping leveraging hierarchical representations," 2022. IEEE RA-L 2023.

[8] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," 2023. ICRA 2024.

[9] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3D scene graphs for language-grounded robot navigation," 2024. Robotics: Science and Systems (RSS).

[10] buildingSMART International, "Industry Foundation Classes (IFC4) – ISO 16739-1:2018." https://technical.buildingsmart.org/standards/ifc/, 2018.

[11] U. V. B. L. Udugama, G. Vosselman, and F. Nex, "Mono-hydra: Real-time 3D scene graph construction from monocular camera input with IMU," 2023.

[12] H. Bavle, J. L. Sanchez-Lopez, M. Shaheer, J. Civera, and H. Voos, "S-Graphs 2.0 – a hierarchical-semantic optimization and loop closure for SLAM," 2025.

[13] M. Shaheer, H. Bavle, J. L. Sanchez-Lopez, and H. Voos, "Robot localization using situational graphs and building architectural plans," 2022.

[14] M. Shaheer, J. A. Millan-Romera, H. Bavle, J. L. Sanchez-Lopez, J. Civera, and H. Voos, "Graph-based global robot simultaneous localization and mapping using architectural plans," 2023.

[15] N. M. Ndulue, J. A. Millan-Romera, M. Giorgi, H. Voos, and J. L. Sanchez-Lopez, "Learning-based hierarchical scene graph matching for robot localization leveraging prior maps," 2026.

[16] J. Wald, H. Dhamo, N. Navab, and F. Tombari, "Learning 3D semantic scene graphs from 3D indoor reconstructions," 2020. CVPR 2020.

[17] S.-C. Wu, J. Wald, K. Tateno, N. Navab, and F. Tombari, "SceneGraphFusion: Incremental 3D scene graph prediction from RGB-D sequences," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7515–7525, 2021.

[18] S. Koch, N. Vaskevicius, M. Colosi, P. Hermosilla, and T. Ropinski, "Open3DSG: Open-vocabulary 3D scene graphs from point clouds with queryable objects and open-set relationships," 2024. CVPR 2024.

[19] D. Maggio, Y. Chang, N. Hughes, M. Trang, D. Griffith, C. Dougherty, E. Cristofalo, L. Schmid, and L. Carlone, "Clio: Real-time task-driven open-set 3D scene graphs," 2024.

[20] X. Hu, C. Zhang, A. Delitzas, X. Zhang, M. Pollefeys, F. Engelmann, and X. Ji, "Hierarchical and holistic open-vocabulary functional 3D scene graphs for indoor spaces," 2026.

[21] W. Xu, V. Ila, L. Zhou, and C. T. Jin, "TB-HSU: Hierarchical 3D scene understanding with contextual affordances," 2024. AAAI 2025.

[22] A. Werby, D. Rotondi, F. Scaparro, and K. O. Arras, "KeySG: Hierarchical keyframe-based 3D scene graphs," 2025.

[23] T. Zemskova and D. Yudin, "3DGraphLLM: Combining semantic graphs and large language models for 3D scene understanding," 2024.

[24] H. Yin, X. Xu, Z. Wu, J. Zhou, and J. Lu, "SG-Nav: Online 3D scene graph prompting for LLM-based zero-shot object navigation," 2024. NeurIPS 2024.

[25] X. Huang, S. Zhao, Y. Wang, X. Lu, W. Zhang, R. Qu, W. Li, Y. Wang, and C. Wen, "MSG-Nav: Unleashing the power of multi-modal 3D scene graph for zero-shot embodied navigation," 2025. CVPR 2026.

[26] J. Strader, A. Ray, J. Arkin, M. B. Peterson, Y. Chang, N. Hughes, C. Bradley, Y. X. Jia, C. Nieto-Granda, R. Talak, C. Fan, L. Carlone, J. P. How, and N. Roy, "Language-grounded hierarchical planning and execution with multi-robot 3D scene graphs," 2025.

[27] P. Pauwels and W. Terkaj, "EXPRESS to OWL for construction industry: Towards a recommendable and usable ifcOWL ontology," Automation in Construction, vol. 63, pp. 100–133, 2016.

[28] M. H. Rasmussen, M. Lefrançois, G. F. Schneider, and P. Pauwels, "BOT: The building topology ontology of the W3C linked building data group," Semantic Web, vol. 12, no. 1, pp. 143–161, 2021.

[29] S. Iranmanesh, H. Saadany, and E. Vakaj, "LLM-assisted Graph-RAG information extraction from IFC data," 2025.

[30] C. Du, S. Esser, S. Nousias, and A. Borrmann, "Text2BIM: Generating building models using a large language model-based multi-agent framework," 2024.

[31] B. K. Nithyanantham, C. Kujat, T. Sesterhenn, S. Telgmann, A. Nedungadi, J. Plönnigs, C. Bartelt, and S. Lüdtke, "BIM-Edit: Benchmarking large language models for IFC-based building information modeling," 2026.

[32] J. Zhu, M. O. Wong, N. Nisbet, J. Xu, T. Kelly, S. Zlatanova, and I. Brilakis, "Semantics-based connectivity graph for indoor pathfinding powered by IFC-Graph," Automation in Construction, vol. 171, p. 106019, 2025.

[33] M. A. Gopee, S. A. Prieto, and B. García de Soto, "Improving autonomous robotic navigation using IFC files," Construction Robotics, vol. 7, no. 3-4, pp. 235–251, 2023.

[34] S. Karimi, I. Iordanova, and D. St-Onge, "An ontology-based approach to data exchanges for robot navigation on construction sites," 2021.

[35] M. A. V. Torres, A. Braun, and A. Borrmann, "Occupancy grid map to pose graph-based map: Robust BIM-based 2D-LiDAR localization for lifelong indoor navigation in changing and dynamic environments," 2023.

[36] A. Longo, C. Chung, M. Palieri, S.-K. Kim, A. Agha, C. Guaragnella, and S. Khattak, "Pixels-to-graph: Real-time integration of building information models and scene graphs for semantic-geometric human-robot understanding," 2025. IEEE CASE 2025.

[37] G. N. Lilis, G. I. Giannakis, and D. V. Rovas, "Automatic generation of second-level space boundary topology from IFC geometry inputs," Automation in Construction, vol. 76, pp. 108–124, 2017.

[38] E. Fichter, V. Richter, J. Frisch, and C. van Treeck, "Automatic generation of second level space boundary geometry from IFC models," in Proceedings of Building Simulation 2021: 17th Conference of IBPSA, 2021.

[39] R. Bormann, F. Jordan, W. Li, J. Hampp, and M. Hägele, "Room segmentation: Survey, implementation, and analysis," in 2016 IEEE International Conference on Robotics and Automation (ICRA), pp. 1019–1026, 2016.

[40] S. Ochmann, R. Vock, R. Wessel, and R. Klein, "Automatic reconstruction of parametric building models from indoor point clouds," Computers & Graphics, vol. 54, pp. 94–103, 2016.

[41] T. Kim and B.-C. Min, "Semantic layering in room segmentation via LLMs," 2024.

A Design Breakdown: Reverse-Engineering the Pipelines

This appendix decomposes the most relevant systems into their constituent design decisions, states the context in which each decision is sound, and records whether it transfers to the BIM setting.

A.1 The canonical layered architecture

Essentially every system in §2 instantiates the same skeleton. Figure 1 shows it alongside the BIM-Graphs variant.

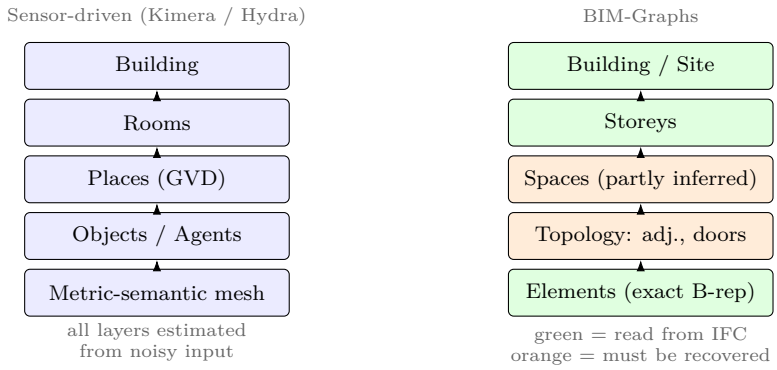


Figure 1: The shared layered skeleton. In sensor-driven systems every layer is estimated; in BIM-Graphs the outer layers are exact and the middle of the hierarchy — exactly the part robots need — is what goes missing.

Why the inversion matters. Sensor pipelines are most confident at the top (a building is a building) and least confident at the bottom (is that a chair?). IFC inverts this: element geometry and class are authoritative, while the room layer is unreliable or absent. Design effort must therefore be spent in the opposite place from where the robotics literature spends it.

A.2 Component-level breakdown

Component	Origin	Why it works there	Transfer to BIM-Graphs
GVD / Voronoi place graph	Kimera [2], Hydra [5], HOV-SG [9]	Free space is the only reliable signal from an occupancy map; doorway constrictions appear as Voronoi bottlenecks, giving room cuts for free.	Transfers, with a better input. We can rasterise exact free space from IFC solids instead of a noisy map, so the "pinch point" heuristic becomes near-exact. Primary tool for storeys lacking IfcSpace.
Rooms as optimisation factors	S-Graphs [7, 12]	A room constrains four wall planes, so asserting it reduces pose drift; the hierarchy also permits marginalising redundant keyframes.	Does not transfer as stated — there is no pose to optimise. The hierarchical marginalisation idea does transfer as a query-time optimisation for large graphs.
Loop-closure-aware hierarchy	Hydra [5]	Deforming the whole graph on loop closure avoids costly rebuilds.	Not applicable. No drift, no loop closure. This is precisely the complexity BIM lets us delete.
Volumetric walls + graph-cut room labelling	Ochmann [40]	Enforces that walls are solids shared between two rooms, keeping topology consistent and resolving double-sided surfaces.	Transfers directly, and answers the "double-sided wall" issue named in our specification. IFC already gives thickness, so the hard estimation step is skipped.
2nd-level space boundary generation	Lilis [37], Fichter [38]	Energy simulation needs surface-exact room boundaries; the methods are built assuming IFC data is broken.	Core import. This is the closest existing solution to our IfcRelSpaceBoundary=0 problem and is under-cited in robotics.
Door-mediated space connectivity	IFC-Graph [32], IndoorGML-style models	Doors are the only legitimate inter-room transitions, so the navigation graph is spaces-as-nodes, doors-as-edges.	Transfers, but must be earned. Without space boundaries we must resolve each door's two adjoining rooms geometrically, by casting along the door normal.
Open-vocab. features per node	ConceptGraphs [8], Open3DSG [18]	Recovers semantics a closed detector misses; CLIP embeddings make nodes queryable by text.	Partly redundant. IFC already names elements and labels 149/155 rooms. Worth reserving for affordances, which IFC never encodes.
Task-driven pruning	Clio [19]	Task-agnostic graphs are needlessly large; an information-bottleneck criterion keeps only task-relevant granularity.	Highly relevant. A single IFC yields 10^4–10^5 entities, most of them irrelevant (individual curtain-wall mullions). Pruning is a necessity.
Collapse/expand LLM interface	SayPlan [4]	Keeps building-scale graphs inside a context window by exposing one subgraph at a time.	Transfers directly and is the intended shape of our QnA interface.
Text-to-Cypher Graph-RAG	[29]	Grounds answers in retrieved triples rather than parametric memory, reducing hallucination.	Adopted by the design specification; complements rather than replaces collapse/expand.

Table 3: Reverse-engineered components, the context that justifies each, and whether it survives the move from sensor data to BIM.

A.3 The recovery pipeline implied by the analysis

Combining the transferable components yields the design in Figure 2: read what IFC states, then recover what it omits, and only then attach open-vocabulary semantics.

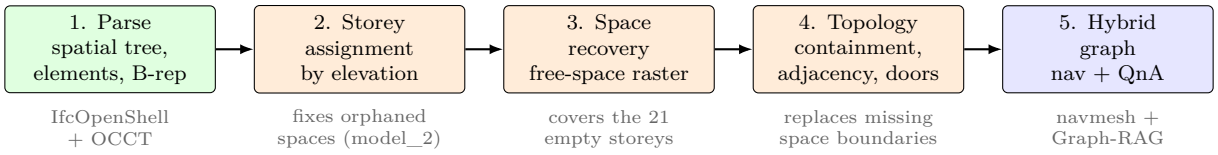


Figure 2: The pipeline implied by the gap analysis. Green is read from IFC; orange is recovered; blue is the deliverable. Stages 2–4 are the research contribution and are exactly the stages the original design specification assumed away.

A.4 Consequences for the original design specification

The reverse-engineering exercise suggests three amendments to Hybrid_BIM2Graph_Design_Spec:

1. The traversal IfcProject→IfcBuildingStorey→IfcSpace→IfcProduct is not viable as the primary path. It yields nothing for 21 of 35 storeys and nothing at all for model_2. It should be demoted to a fast path with geometric recovery as the guaranteed fallback.
2. Double-sided wall handling should follow Ochmann [40] rather than normal-duplication. Duplicating boundaries along the thickness vector re-derives, less robustly, a volumetric representation IFC already provides explicitly.
3. Stage 2/3 rendering is not on the critical path. Because IFC labels 671/681 rooms and names every element, a metrically exact and topologically complete graph is attainable with no rendering whatsoever. VLM injection should be scoped to affordances — the one thing IFC genuinely lacks — rather than treated as a prerequisite for graph construction. The implementation in §7 confirms this: the full graph, including navigation and QnA interfaces, was produced without rendering a single frame.

B Annotation Protocol for Ground Truth

§7.4 identifies annotation as the binding constraint. Because annotating 25 buildings from scratch is prohibitive, we annotate by correction: the pipeline's prediction is displayed over the true floor plan and the annotator adjudicates it. This converts an open-ended drawing task into a sequence of bounded judgements, and yields exactly the quantities the necessary conditions cannot supply.

B.1 What is annotated

1. Room verdicts. Each predicted room is marked correct, spurious (not a real room — a shaft, void or artefact), merge (two predictions that are one room) or split (one prediction covering several). A corrected functional label may be supplied, which also gives ground truth for semantic labelling.
2. Missing rooms. A real room the pipeline failed to find is marked by placing a point inside it and labelling it. A point suffices for recall; a polygon is optional, since exact boundaries are rarely in dispute.
3. Connection verdicts. Each predicted connected_by_door, open_passage and vertically_connected edge is confirmed or rejected, and missing connections are added by selecting two rooms. This is the critical annotation: connectivity is what the schema omits entirely and what navigation depends on.

B.2 Metrics it enables

Room precision/recall and connectivity precision/recall follow directly, both reportable separately for rooms stated by IFC versus rooms recovered geometrically — the comparison that isolates the contribution of the recovery stage from what the file already provided. Label accuracy over corrected functional names measures the semantic layer. Together these replace the necessary conditions of Table 2 with genuine accuracy.

B.3 Design constraints

Three properties matter for a tool that colleagues will actually use. It must open an IFC and show a floor plan in seconds, so geometry is pre-extracted and cached rather than tessellated on demand. Every judgement must be one click, with keyboard shortcuts for the common verdicts. And annotations must be stored per annotator, so that the same model can be labelled independently by two people and inter-annotator agreement measured — without which

we cannot know whether disputed corridors reflect pipeline error or genuine ambiguity about where one room ends and the next begins.